# Predictive HW2
# Dogs vs Cats Image Classification

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## Data Preprocessing

- Downloaded and extracted 25,000 training images and 12,500 test images
- Organized training images into separate `cats/` and `dogs/` subfolders (required by ImageDataGenerator)
- Resized all images to **224×224** pixels
- Applied **ImageNet normalization** using model-specific preprocessing functions
- Split training data into **80% train / 20% validation** using `validation_split=0.2`

## Data Augmentation

Applied the following augmentation techniques to training images to improve generalization:

- Horizontal and vertical flip
- Rotation up to 20 degrees
- Zoom range of 0.2
- Width and height shift of 0.2
- Shear range of 0.2

## Model Exploration

Two models were trained and compared:

### Model 1 — CNN from Scratch (Baseline)

- 3 convolutional blocks (32, 64, 128 filters)
- MaxPooling after each block
- Dense(128) + Dropout(0.5) + Sigmoid output
- Optimizer: Adam
- Loss: Binary Crossentropy
- Epochs: 6 with Early Stopping
- **Best Validation Accuracy: ~77%**

### Model 2 — EfficientNetB4 (Transfer Learning)

- Pretrained on ImageNet

- Added GlobalAveragePooling2D + BatchNormalization + Dense(512) + Dropout(0.5) + Dense(256) + Dropout(0.3) + Sigmoid
- Initial training: frozen base, lr=1e-3, 10 epochs
- Fine tuning: unfreeze last 30 layers, lr=1e-5, 6 epochs
- **Best Validation Accuracy: ~99%**
- **Kaggle Score: 0.06584**

**Model 3 — ResNet50 (Transfer Learning)**

- Pretrained on ImageNet
- Added GlobalAveragePooling2D + BatchNormalization + Dense(512) + Dropout(0.5) + Dense(256) + Dropout(0.3) + Sigmoid
- Image size: 256×256
- Initial training: frozen base, lr=1e-3, 10 epochs
- Fine tuning: unfreeze last 30 layers, lr=1e-5, 4 epochs
- **Best Validation Accuracy: ~98%**
- **Kaggle Score: 0.05453**

## Overfitting Prevention

- **Dropout** (0.5 and 0.3) in fully connected layers
- **BatchNormalization** between dense layers
- **Early Stopping** with patience=3
- **Data Augmentation** during training

## Validation Strategy

- 80/20 train/validation split
- Monitored validation loss with Early Stopping
- Restored best weights after training

## Key Findings

- Transfer learning significantly outperforms CNN from scratch
- ResNet50(with clipping) outperformed EfficientNetB4(without clipping - thus Final Consideration) on this dataset. EfficientNet (with clipping - gave a score of 0.04553)
- BatchNormalization and more augmentation improved model generalization
- Fine tuning pretrained layers gave better results than freezing them.

**Screen Shot of the score on Kaggle -**

